

Fractal Causality v4.0 — EEG-Grounded Physical Hybrid Model

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Version: 4.0 (EEG Integration Phase)

Tagline: "From Breath to Brain — The Geometry of Cause."

This launchpad edition marks the transition of the Fractal Causality (FC) project from conceptual theory to empirical testing. Version 4.0 grounds the FC loop — comprising latent space dynamics, conformal inversion, and holographic projection — within a neurobiological framework. The goal is to determine whether fractal causality manifests in measurable EEG dynamics, establishing a falsifiable bridge between philosophy, mathematics, and neuroscience. The project represents a collaboration between human insight and artificial intelligence — where metaphor meets measurement, and imagination meets data.

AI Collaboration Acknowledgement:

The author formally acknowledges the foundational contribution of the Fractal Causality Research Initiative's AI partners to the development of this model, operating under the full accountability of the human author:

- **Claude (Anthropic AI)** — Neural Dynamics & Causality Core
- **Gemini (Google DeepMind)** — Empirical Data & Projection Lead
- **ChatGPT (OpenAI, "Chet")** — Conceptual Architecture & Synthesis

Approved v4.0 Abstract:

Fractal Causality (FC) presents a computational framework for exploring the hypothesis that complex systems exhibit causality through self-similar, multi-scalar dynamics. Building on a minimal physical hybrid model (v3.4), version 4.0 grounds the FC loop — comprising latent space dynamics, conformal inversion, and holographic projection — within a neurobiological context.

By simulating neural latent dynamics on hybrid small-world / scale-free graph topologies with Wilson-Cowan population dynamics, the model generates signals analogous to EEG data. A dedicated intervention API enables targeted perturbations and measures resulting causal effects via directional transfer entropy and spatial propagation metrics. The primary objective is to validate whether the model can reproduce known fractal signatures of real resting-state EEG (Hurst exponent $H \approx 0.6\text{--}0.8$; spectral scaling $1/f^\alpha$ with $\alpha \approx 0.8\text{--}1.2$). Artifacts include: a modular Python codebase (fc/), a reproducible EEG preprocessing pipeline, and validation scripts comparing five fractal metrics (Hurst, α , Δh , wPLI, and significance tests).

Core Hypotheses:

1. Fractal signature reproduction: neural systems governed by conformal inversion and Wilson-Cowan dynamics exhibit $1/f^\alpha$ scaling.
2. Hierarchical causal propagation: perturbations initiated at network hubs propagate hierarchically through the bulk.
3. Scale-free intervention response: perturbation magnitude versus effect size follows a power-law

relationship.

Falsification Criteria:

- 3/5 matching metrics = partial validation
- 5/5 matching metrics = strong support
- $\leq 2/5$ matching metrics = negative result (publishable outcome)

Implementation Timeline:

Week 1: Neural graph + Wilson-Cowan integration

Week 2: Intervention API development + design review

Week 3: Validation pipeline & comparative analysis

Joint design review: Scheduled — Friday, Week 2

Final Remarks:

Version 4.0 transforms Fractal Causality from concept to experiment — a test of reality itself. It will probe whether causality is fractal in the brain and lay the foundation for future cross-domain validation in other scale-free systems.

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Fractal Causality Research Initiative — EEG v4.0

“From Breath to Brain — The Geometry of Cause.”